Sunkari Rama Harshini

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	5.65
Sasi Educational institute	Class XII (Board of Intermediate Education)	2023	93.80%
Sri Chaitanya School	Class X (Board of Secondary Education)	2021	99.50%

SCHOLASTIC ACHIEVEMENTS

SSC Intermediate - 938/1000

SSC State Borad Exams - 597/600

POSITIONS OF RESPONSIBILITY

- Worked as the Hostel Representative for the block, facilitating communication between residents and management.

SKILLS

Languages : C, python, SQL

ML frameworks : PyTorch

Python libraries : numpy, pandas, matplotlib

Tools : Git, Github, Docker, VScode

RELEVANT COURSES

Artificial Intelligence : Introduction to Modren AI, Programming for AI, Probability and Random Variables, Foundations of Machine Learning, Convex Optimization, Deep Learning, Reinforcement Learning, Robotics, Computer Vision, Natural Language Processing.

Electrical Engineering : Vector Calculus, Matrix Theory, Linear Systems and Signal Processing, Concentration Inequalities.

Computer Science : Introduction to Programming, Discrete Maths, Data structures, Algorithms, Compilers-I, DBMS-I,II, Data Mining.

Math : Calculus, Differential Equations, Series of functions, Metric spaces, Applied statistics, Introduction to Bayesian Statistics, Predictive Modelling.

PROJECTS

Manipulation Detection System(Team project) :

- Built an NLP system for detecting 11 manipulation techniques in conversational data.
- Generated sentence embeddings using Sentence Transformers (all-mpnet-base-v2).
- Developed multi-class and multi-label classifiers using SVM, MLP, and ensemble learning methods.
- Enhanced chatbot safety by adapting responses based on detected manipulation patterns.

Deep Learning(Link) :

- Neural Network Fundamentals & Optimization: Implemented machine learning fundamentals including convex optimization, L1/L2 regularization analysis, PCA-based feature extraction, and sigmoid neuron training using Python and NumPy.
- Feed-Forward Neural Network: Developed a feed-forward neural network from scratch with backpropagation and gradient-based learning, achieving image classification through NumPy-based implementations.
- RNN/LSTM/GRU Project: Built and evaluated RNN, LSTM, and GRU architectures for sequence prediction, comparing recurrent models through loss analysis and performance visualization.

Computer Vision Image Processing & Restoration (Link) :

- Implemented from-scratch image-processing pipelines for gamma correction, histogram equalization, 2D convolution/correlation, and zero/reflect/replicate padding with quantitative color and boundary-artifact analysis.
- Compared box, Gaussian, median, and bilateral filtering across multiple kernel sizes; evaluated denoising using PSNR, edge-preservation loss, and a combined quality score.
- Built SVD-based separable filtering and analyzed speed-accuracy trade-offs of rank-1 approximations for non-separable kernels.
- Developed FFT-based periodic-noise removal using manually designed and automatically detected notch-reject filters, and generated frequency-domain hybrid images.

Machine Learning Projects (Link) :

- MNIST data classification using KNN. Experimented with different types of kernels, regularisers to approximate functions using regression.
- Developed different types of classification models like least squares approach, linear discriminant analysis, fisher analysis,perceptron ,logistic regression.
- Estimated parameters of Gaussian Mixture Models (GMM) using Expectation-Maximization (EM) algorithm.
- Built a neural network from scratch for MNIST data classification. Compared multiple activation functions, optimizers, regularization methods to get the most effective model

EXTRACURRICULAR

- I am very much interested in chess and I won Silver medal in Diesta - 2024 and Bronze Medal in our chess league - 2024

EXPERIENCE

Diagonal Matrix Ltd. -- AI Intern

- Contributed to JAI, a Langflow-based AI platform, working across AI agents, agentic workflows, LLM integrations, and AI accelerator development.
- Developed and integrated RAG-based AI workflows, APIs, and agent components for building and executing LLM-powered applications.
- Worked with FastAPI, React/Vite, Git/GitHub, Docker, MCP, and REST/SSE APIs across backend and frontend development workflows.
- Contributed to Agent Workspace and Agentic Assistant functionality, including session management, API integration, streaming workflows, validation, and retry mechanisms.
- Used Docker-based development and packaging workflows and Git/GitHub for collaborative version control and engineering practices.